Name: ____________________________ Date: ________________ Period: _________

Primary Objective

I will be able to...

1. **LO 4.2.A** — Identify key characteristics of a parametric planar motion function related to position: find the horizontal and vertical extrema from the max/min values of $x(t)$ and $y(t)$, and locate x- and y-intercepts of the curve from the zeros of $y(t)$ and $x(t)$. ____

Vocabulary & Key Concepts

horizontal extremum: The _______________ or _______________ value of $x(t)$; gives the rightmost or leftmost position of the particle.

vertical extremum: The maximum or minimum value of ___________; gives the highest or lowest position of the particle.

y-intercept (of parametric curve): A point on the curve where $x(t) =$ ______; found by solving $x(t) = 0$ for t, then evaluating $y(t)$ at those values.

x-intercept (of parametric curve): A point on the curve where $y(t) =$ ______; found by solving $y(t) = 0$ for t, then evaluating $x(t)$.

Hook

Launch: A ball's position (feet) at time t seconds is $x(t) = -16t^2 + 48t$ and $y(t) = 12t - t^2$.

Q1. Which component do you examine to find how high the ball goes? What equation do you solve to find when the ball lands? ____

____

LO 4.2.A — Planar Motion: Horizontal and Vertical Extrema

Example: Let $f(t) = (x(t), y(t)) = (t^2 - 4t + 3,\ -t^2 + 2t)$ for $0 \le t \le 4$.

Step 1 — Build the table.

t	$x(t) = t^2 - 4t + 3$	$y(t) = -t^2 + 2t$	$(x,\ y)$
0	______________	______________	________________
1	______________	______________	________________
2	______________	______________	________________
3	______________	______________	________________
4	______________	______________	________________

Step 2 — Find horizontal extrema of $x(t) = t^2 - 4t + 3$.

$x(t)$ is a _____________ parabola. Vertex at $t = -\dfrac{b}{2a} = -\dfrac{}{} =$ _______

Minimum value of x: $x(\_\_\_\_\_) = \_\_\_\_\_\_\_\_\_\_\_\_$ (particle's _____________-most position)
Maximum value of x on $[0, 4]$: check endpoints. $x(0) = \_\_\_\_\_\_\_$, $x(4) = \_\_\_\_\_\_\_$. Maximum is _______ (particle's _____________-most position).

Step 3 — Find vertical extrema of $y(t) = -t^2 + 2t$.
Vertex at $t = -\dfrac{b}{2a} = \_\_\_\_\_\_\_$.
Maximum value of y: $y(\_\_\_\_\_) = \_\_\_\_\_\_\_\_\_\_\_\_$ (particle's ___________-most position)
Minimum of y on $[0, 4]$: check endpoints. $y(0) = \_\_\_\_\_\_\_$, $y(4) = \_\_\_\_\_\_\_$. Minimum is _______.

LO 4.2.A — Axis Intercepts of a Parametric Curve

Key rule: Zeros of $x(t)$ give _________-intercepts of the curve. Zeros of $y(t)$ give _________-intercepts.

Example (continued): $f(t) = (t^2 - 4t + 3,\ -t^2 + 2t)$, $0 \leq t \leq 4$.

Find the y-intercepts (where $x(t) = 0$):
Set $t^2 - 4t + 3 = 0$. Factor: $(t - \_\_\_\_\_)(t - \_\_\_\_\_) = 0$.
$t = \_\_\_\_\_\_$ or $t = \_\_\_\_\_\_$.
Evaluate y at each: $y(\_\_\_\_) = \_\_\_\_\_\_\_$, $y(\_\_\_\_) = \_\_\_\_\_\_\_$.
y-intercepts of the curve: __________________ and __________________.

Find the x-intercepts (where $y(t) = 0$):
Set $-t^2 + 2t = 0$. Factor: $-t(t - \_\_\_\_\_) = 0$.
$t = \_\_\_\_\_\_$ or $t = \_\_\_\_\_\_$.
Evaluate x at each: $x(\_\_\_\_) = \_\_\_\_\_\_\_$, $x(\_\_\_\_) = \_\_\_\_\_\_\_$.
x-intercepts of the curve: __________________ and __________________.